

## NORFOLK NAS, VA, USA

WMO: 723085

Lat: 36.937N

Lon: 76.289W

Elev: 5

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13750

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.3       | -2.9 | -16.4                         | 0.9 | -3.0 | -13.7 | 1.1 | -0.8 | 12.7                  | 2.5  | 11.6 | 2.7  | 5.6                   | 330  | 0.474 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.8                    | 34.4            | 25.0 | 33.1 | 24.5 | 32.1 | 24.2 | 26.5                | 31.9 | 25.8 | 30.9 | 25.3 | 30.1 | 3.9                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.0                            | 20.2 | 28.9 | 24.2 | 19.1 | 28.3 | 23.8 | 18.7 | 28.0 | 82.3          | 32.0 | 79.5 | 31.4 | 77.0 | 30.0 | 30.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| 11.2              | 9.6  | 8.5 | DB                         | -7.9 | 37.0               | 2.9 | 1.3                                                | -9.9  | 38.0       | -11.6 | 38.7       | -13.2 | 39.4       | -15.3 | 40.4 | (4) |
|                   |      |     | WB                         | -9.7 | 27.9               | 2.6 | 1.0                                                | -11.6 | 28.7       | -13.1 | 29.3       | -14.5 | 29.8       | -16.4 | 30.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 16.7   | 5.8   | 7.0   | 10.5  | 16.0  | 20.6  | 25.0  | 27.1  | 26.3  | 23.6  | 18.0  | 12.0  | 8.3  |
|                                                                               | DBStd       | 8.66   | 5.76  | 5.13  | 5.21  | 4.78  | 4.11  | 3.19  | 2.43  | 2.32  | 2.79  | 4.14  | 4.55  | 5.04 |
|                                                                               | HDD10.0     | 457    | 158   | 112   | 59    | 6     | 0     | 0     | 0     | 0     | 0     | 1     | 29    | 94   |
|                                                                               | HDD18.3     | 1652   | 391   | 318   | 249   | 102   | 25    | 1     | 0     | 0     | 1     | 58    | 194   | 313  |
|                                                                               | CDD10.0     | 2915   | 26    | 28    | 73    | 185   | 328   | 451   | 532   | 507   | 407   | 248   | 90    | 41   |
|                                                                               | CDD18.3     | 1070   | 1     | 1     | 6     | 32    | 95    | 203   | 273   | 248   | 158   | 47    | 5     | 2    |
|                                                                               | CDH23.3     | 9283   | 1     | 6     | 40    | 253   | 722   | 1835  | 2837  | 2311  | 1008  | 248   | 17    | 5    |
|                                                                               | CDH26.7     | 3292   | 0     | 0     | 4     | 67    | 229   | 698   | 1164  | 819   | 262   | 48    | 1     | 0    |
| Wind                                                                          | WSAvg       | 4.1    | 4.5   | 4.5   | 4.8   | 4.6   | 4.0   | 3.6   | 3.4   | 3.3   | 4.0   | 4.0   | 4.1   | 4.1  |
| Precipitation                                                                 | PrecAvg     | 1045   | 86    | 84    | 90    | 77    | 77    | 89    | 114   | 111   | 102   | 74    | 65    | 76   |
|                                                                               | PrecMax     | 1722   | 226   | 206   | 230   | 169   | 236   | 244   | 309   | 300   | 463   | 242   | 160   | 167  |
|                                                                               | PrecMin     | 553    | 22    | 24    | 21    | 12    | 16    | 21    | 33    | 24    | 13    | 5     | 2     | 12   |
|                                                                               | PrecStd     | 268    | 38    | 39    | 42    | 42    | 43    | 48    | 59    | 67    | 105   | 51    | 41    | 40   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 22.6  | 24.0  | 27.3  | 31.2  | 33.0  | 35.9  | 36.4  | 35.5  | 33.1  | 30.0  | 25.5  | 23.9 |
|                                                                               |             | MCWB   | 17.3  | 17.4  | 18.0  | 20.2  | 23.1  | 24.6  | 25.5  | 25.3  | 23.6  | 22.2  | 18.8  | 18.6 |
|                                                                               | 2%          | DB     | 20.0  | 20.9  | 24.0  | 28.6  | 31.1  | 33.7  | 34.6  | 33.6  | 31.2  | 28.0  | 23.0  | 21.2 |
|                                                                               |             | MCWB   | 15.9  | 15.1  | 16.4  | 19.0  | 22.3  | 24.1  | 25.5  | 25.1  | 23.4  | 21.4  | 18.3  | 17.4 |
|                                                                               | 5%          | DB     | 17.6  | 18.3  | 21.8  | 26.5  | 29.2  | 32.2  | 33.3  | 32.3  | 29.5  | 26.2  | 21.2  | 18.8 |
|                                                                               |             | MCWB   | 14.3  | 13.7  | 15.4  | 18.2  | 21.6  | 23.8  | 25.1  | 24.5  | 22.9  | 20.8  | 17.1  | 16.0 |
|                                                                               | 10%         | DB     | 15.0  | 15.8  | 19.2  | 23.9  | 27.4  | 30.7  | 32.1  | 30.9  | 27.9  | 24.2  | 19.1  | 16.5 |
|                                                                               |             | MCWB   | 11.7  | 11.4  | 14.3  | 17.4  | 20.7  | 23.2  | 24.8  | 24.2  | 22.5  | 19.8  | 15.7  | 13.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 18.7  | 18.5  | 19.5  | 21.9  | 25.0  | 26.6  | 27.9  | 27.0  | 25.6  | 24.4  | 21.0  | 20.0 |
|                                                                               |             | MCDB   | 21.6  | 22.4  | 24.6  | 27.7  | 30.1  | 32.1  | 32.0  | 32.3  | 29.7  | 27.5  | 23.5  | 22.7 |
|                                                                               | 2%          | WB     | 16.7  | 16.7  | 18.1  | 20.7  | 23.6  | 25.5  | 26.8  | 26.1  | 24.9  | 22.9  | 19.4  | 17.9 |
|                                                                               |             | MCDB   | 19.2  | 20.0  | 22.1  | 26.4  | 28.9  | 31.2  | 32.1  | 31.3  | 28.5  | 26.1  | 21.9  | 20.4 |
|                                                                               | 5%          | WB     | 14.6  | 14.6  | 16.7  | 19.6  | 22.6  | 24.8  | 26.1  | 25.6  | 24.3  | 21.7  | 17.8  | 16.3 |
|                                                                               |             | MCDB   | 17.5  | 17.9  | 20.7  | 24.8  | 27.6  | 30.1  | 31.6  | 30.3  | 27.6  | 25.0  | 20.3  | 18.6 |
|                                                                               | 10%         | WB     | 12.0  | 11.7  | 14.8  | 18.2  | 21.6  | 24.2  | 25.5  | 25.0  | 23.7  | 20.5  | 16.2  | 13.9 |
|                                                                               |             | MCDB   | 14.6  | 15.2  | 18.8  | 22.9  | 26.2  | 28.9  | 30.6  | 29.4  | 26.8  | 23.8  | 18.9  | 16.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 8.0   | 8.6   | 9.0   | 9.7   | 8.7   | 8.1   | 7.8   | 7.3   | 6.9   | 7.9   | 8.2   | 7.8  |
|                                                                               |             | MCDBR  | 11.8  | 13.0  | 13.0  | 12.7  | 11.0  | 10.2  | 9.6   | 9.1   | 9.0   | 9.6   | 10.6  | 10.5 |
|                                                                               |             | MCWBR  | 9.0   | 9.1   | 7.9   | 6.2   | 5.2   | 4.0   | 3.6   | 3.5   | 3.9   | 4.9   | 6.9   | 8.2  |
|                                                                               | 5% WB       | MCDBR  | 11.2  | 12.2  | 11.6  | 11.4  | 9.6   | 9.0   | 8.7   | 8.0   | 7.1   | 7.5   | 9.0   | 10.2 |
|                                                                               |             | MCWBR  | 9.4   | 9.9   | 8.6   | 6.6   | 5.1   | 4.0   | 4.0   | 3.6   | 3.7   | 4.6   | 7.1   | 8.7  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.308  | 0.321 | 0.347 | 0.391 | 0.434 | 0.483 | 0.511 | 0.505 | 0.407 | 0.356 | 0.324 | 0.313 |      |
|                                                                               | taud        | 2.540  | 2.501 | 2.457 | 2.347 | 2.257 | 2.171 | 2.109 | 2.112 | 2.376 | 2.513 | 2.550 | 2.559 |      |
|                                                                               | Ebn at Noon | 888    | 918   | 919   | 891   | 853   | 808   | 782   | 778   | 845   | 866   | 861   | 859   |      |
|                                                                               | Edh at Noon | 78     | 91    | 104   | 123   | 137   | 149   | 157   | 154   | 112   | 89    | 77    | 72    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.34   | 3.10  | 4.14  | 5.28  | 5.94  | 6.38  | 6.08  | 5.48  | 4.46  | 3.54  | 2.65  | 2.09  |      |
|                                                                               | RadStd      | 0.19   | 0.33  | 0.33  | 0.38  | 0.51  | 0.41  | 0.30  | 0.31  | 0.36  | 0.40  | 0.21  | 0.17  |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            |  | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +150    | +96     |
| Regional (57 neighbors) |  | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +166    | +106    |

Nomenclature: See separate page